



PERTH MODERN SCHOOL
Exceptional schooling. Exceptional students.

Test Two

Semester One 2017 ***UNIT 1 METHODS***

Calculator Free 35 minutes /30 marks

Only Formula Sheet Permitted

Name:

Place a tick in the box next to your Mathematics teachers name:

Mr Strain	☐
Ms Sindel	☐
Ms Rimando	☐
Ms Reynolds	☐
Dr Pearce	☐
Mrs Flynn	☐
Ms Ensly	☐
Mrs Carter	☐

Question 1**(3, 3 = 6 marks)**

Find the equation of each linear function

a) Passing through (2,-3) and (4,1)

b) Perpendicular to the line $2x + y - 3 = 0$ and with x -intercept of -2.

Question 2**(2 marks)**

Given the points $(-3, 1)$ and $(4, 2)$ find the **exact value** of the distance between them.

Question 3**(2 marks)**

The gradient of the straight line between $(3, y)$ and $(-2, 5)$ is $-\frac{3}{5}$. Find the value of y .

Question 4**(1, 1 = 2 marks)**

The quadratic equation $kx^2 + 5x - 3 = 0$ has exactly one real solution.

a) What is the value of the discriminant?

b) Hence, find the value of k .

Question 5**(2, 2 = 4 marks)**

Solve the following quadratic equations giving exact answers

a) $x^2 + 2x - 15 = 0$

b) $x^2 - 3x - 5 = 0$

Question 6**(2, 5 = 7 marks)**

Determine the rules for the following tables

a)

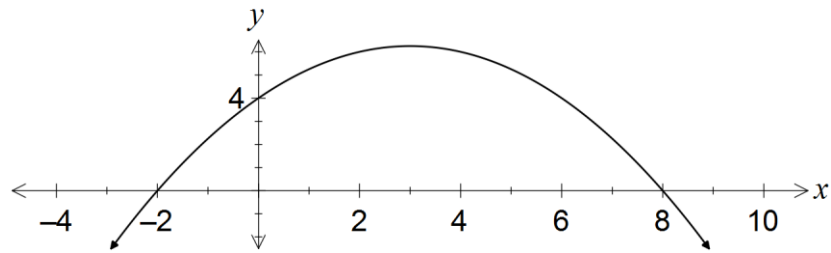
x	-7	-6	-5	-4	-3
y	11	10	9	8	7

b)

x	1	2	3	4	5	6	7
y	2	2	4	8	14	22	32

Question 7**(3, 2, 2 = 7 marks)**

- (a) Part of the graph of $y = ax^2 + bx + 4$ is shown below.



Determine the values of the coefficients a and b .

- (b) A quadratic has equation $y = x^2 - 6x + 2$. Determine

(i) the coordinates of its turning point.

(ii) the exact values of the zeros of the quadratic.